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研究方向

- 双曲守恒律方程高精度数值方法 (WENO 格式等) 研究
- 曲面双曲守恒律方程理论、方法与应用研究
- 多介质流模型高精度保物理约束数值方法研究

教育背景

- 2018.09 – 2022.06 ■ 博士，中国海洋大学数学科学学院计算数学专业
 👤 Wai Sun Don 教授
 📖 《保物理约束的高精度 WENO 格式》
- 2015.09 – 2018.06 ■ 硕士，中国海洋大学数学科学学院计算数学专业
 👤 高振教授，Wai Sun Don 教授
 📖 《基于径向基函数的间断检测算法及其在高精度杂交格式中的应用》
- 2011.08 – 2015.06 ■ 学士，中国海洋大学数学科学学院数学与应用数学专业

工作经历

- 2024.09 – 至今 ■ 青年英才工程副教授，硕导，中国海洋大学数学科学学院
- 2022.08 – 2024.09 ■ 科研博士后，中国海洋大学数学科学学院
 👤 高振教授
 📖 《多介质流体的高精度保物理约束 AWENO 格式》

学术访问

- 2026 ■ 香港浸会大学数学系，2026.01.04–01.15, 2026.09.01–09.19
 👤 Leevan Ling 教授 (数学系主任) 👤 Wai Sun Don 教授
- 2025 ■ 韩国浦项科技大学数学系，访问教授，2025.12.25–2026.01.02
 👤 Jae-Hun Jung 教授 (理学院院长、数学系主任) 👤 Jiaxi Gu 博士
- 2025 ■ 香港浸会大学数学系，2025.11.02–11.29
 👤 Leevan Ling 教授 👤 Wai Sun Don 教授
- 2024 ■ 香港浸会大学数学系，2024.11.03–11.30
 👤 Leevan Ling 教授 👤 Wai Sun Don 教授
- 2024 ■ 南方科技大学杰曼诺夫数学中心，2024.03.03–05.31
 👤 Alexander Kurganov 教授 (讲席教授，入选国家重要人才计划)
- 2024 ■ 香港科技大学数学系，2024.05.05–05.11
 👤 Zhichao Peng 教授

学术访问 (continued)

- 2023  香港浸会大学数学系, 2023.10.16–11.06, 2023.12.16–2024.01.09
 Leevan Ling 教授  Wai Sun Don 教授
- 2023  南方科技大学杰曼诺夫数学中心, 2023.04.03–09
 Alexander Kurganov 教授
- 2019  南方科技大学数学系, 2019.03.27–04.01, 09.15–19, 12.17–20
 Alexander Kurganov 教授

研究成果 (* = 通讯作者; ‡ = 同等贡献; IF = 影响因子; 累积被引 = 486; H-Index = 13)

- 1 Khaled Bensayah, **Bao-Shan Wang***, Jia-Hao Liu, & Abdellah Hadjadj. WENO-ZKB: An improved WENO-Z+ scheme with new anti-dissipative term. *Journal of Computational Physics*, 2027, 569, 115434. (IF = 3.9)
- 2 Changming Guo[‡], **Bao-Shan Wang[‡]**, Wai Sun Don[‡], & Yuanyang Qiao^{*,‡}. High-order physical-constraints-preserving velocity-consistent schemes for compressible multicomponent five-equation model with the Mie-Grüneisen equation of state. *Journal of Computational Physics*, 2026, 563, 115049. (IF = 3.9)
- 3 Ya-Ru Zhao, Zhen Gao, & **Bao-Shan Wang***. Fifth-order equilibrium-preserving path-conservative characteristic-wise AWENO scheme for one-fluid two-temperature Euler model. *Journal of Computational Physics*, 2026, 544, 114409. (IF = 3.9)
- 4 Xiao-Shuo Xiang, Peng Li, Zhen Gao, & **Bao-Shan Wang***. High-order well-balanced positivity-preserving affine-invariant finite volume CWENOZ scheme with LDCU flux for the Euler equations under gravitational fields, *Applied Numerical Mathematics*, 2026, 224, 181–206. (IF = 2.4)
- 5 **Bao-Shan Wang**, Alex Shiu Lun Chu, Leevan Ling, & Wai Sun Don*. Foliation structures and global flow dynamics of scalar hyperbolic conservation laws on manifolds: I. Geometry-compatible fluxes and numerical validation on sphere and torus. *Communications in Nonlinear Science and Numerical Simulation*, 2026, 152, 109415. (IF = 3.8)
- 6 Xiao-Shuo Xiang, Peng Li, & **Bao-Shan Wang***. Finite-volume WENO scheme on non-uniform mesh for 3D detonation simulation. *Applied Mathematics Letters*, 2026, 182, 110046. (IF = 2.7)
- 7 Kang-Bo Tian, **Bao-Shan Wang**, Xiao Wen, & Zhen Gao*. High order well-balanced and positivity-preserving AWENO scheme for rotating shallow water equations with Coriolis force. *East Asian Journal on Applied Mathematics*, 2026, 16(4), 714–741. (IF = 1.1)
- 8 Wai Sun Don[‡], Jia-Le Li[‡], Leevan Ling[‡], **Bao-Shan Wang^{*,‡}**, & Yinghua Wang[‡]. Hybrid high-order shock-capturing scheme for one-dimensional hyperbolic conservation laws on manifolds (surface PDEs) in the time-continuous embedding framework. in: S. Chun, J.-H. Jung, E.J. Park, J. Shen (eds), *Spectral and High-Order Methods for Partial Differential Equations ICOSAHOM 2023. Lecture Notes in Computational Science and Engineering*, vol. 142, Springer, Cham, 2025, pp. 239–255.
- 9 **Bao-Shan Wang** & Wai Sun Don*. Fifth-order bound-, positivity-, and equilibrium-preserving affine-invariant AWENO scheme for two-medium γ -based model of stiffened gas. in: R.S. Myong and H.D. Kim (eds), *Proceedings of the 34th International Symposium on Shock Waves, Volume 1: Fundamentals*, Springer, Singapore, 2025, pp. 59–71.
- 10 Qingcheng Fu, Yaguang Gu*, Alexander Kurganov, & **Bao-Shan Wang**. Bound- and positivity-preserving path-conservative central-upwind AWENO scheme for the five-equation model of compressible two-component flows. *Journal of Scientific Computing*, 2025, 104(3), 94. (IF = 3.3)

- 11 Changming Guo, Yuanyang Qiao, Wai Sun Don, & **Bao-Shan Wang***. A family of bound-preserving velocity-consistent schemes for two-medium γ -based model with stiffened gas. *Journal of Computational Physics*, 2025, 538, 114149. (IF = 3.8)
- 12 Cai-Feng Wang[#], Wai Sun Don[#], Jia-Le Li[#], & **Bao-Shan Wang*[#]**. Improved sixth-order WENO finite difference schemes for hyperbolic conservation laws. *Advances in Applied Mathematics and Mechanics*, 2025, 17(6), 1591–1624. (IF = 1.1)
- 13 Jia-Le Li, Wai Sun Don, Cai-Feng Wang, & **Bao-Shan Wang***. Spatial-temporal adaptive-order positivity-preserving WENO finite difference scheme with relaxed CFL condition for Euler equations with extreme conditions. *Advances in Applied Mathematics and Mechanics*, 2025, 17(3), 804–839. (IF = 1.5)
- 14 Zhen Gao, Zi-Yu Tang, **Bao-Shan Wang***, & Ya-Ru Zhao. Novel high-order alternative finite difference central WENO schemes for hyperbolic conservation laws. *Numerical Methods for Partial Differential Equations*, 2025, 41(2), e23170. (IF = 2.1. 凭此文获 2025 年第一季度 Wiley 威立中国高贡献作者奖)
- 15 Xiao-Shuo Xiang[#], **Bao-Shan Wang[#]**, Zhen Gao[#], & Peng Li^{*[#]}. Affine-invariant WENO operator on nonuniform Cartesian mesh with application to finite volume and discontinuous Galerkin methods. *Journal of Scientific Computing*, 2025, 102(2), 29. (IF = 2.8)
- 16 Baifen Ren, **Bao-Shan Wang**, Xiangxiong Zhang, & Zhen Gao*. Positivity and bound preserving well-balanced high order compact finite difference scheme for Ripa and pollutant transport model. *Computers and Mathematics with Applications*, 2024, 176, 545–563. (IF = 2.9)
- 17 Zhen Gao[#], Shuang Guo[#], **Bao-Shan Wang[#]**, & Yaguang Gu^{*[#]}. High order bound- and positivity-preserving finite difference affine-invariant AWENO scheme for the five-equation model of two-medium flows. *Communications in Computational Physics*, 2024, 36(3), 781–820. (IF = 2.6)
- 18 **Bao-Shan Wang** & Naveen Kumar Garg*. Third-order numerical scheme for Euler equations of gas dynamics using Jordan canonical based splitting flux. *Computers & Fluids*, 2024, 281, 106370. (IF = 2.5)
- 19 Zhen Gao, Ya-Ru Zhao, & **Bao-Shan Wang***. Sixth-order perturbed WENO interpolation-based AWENO and WCNS-E schemes for hyperbolic conservation laws. *Applied Mathematics Letters*, 2024, 158, 109230. (IF = 2.9)
- 20 Qingcheng Fu[#], Zhen Gao[#], Yaguang Gu[#], Peng Li^{*[#]}, & **Bao-Shan Wang[#]**. Improved well-balanced AWENO schemes with hydrostatic reconstruction for the Euler equations under gravitational fields. *Mathematics and Computers in Simulation*, 2024, 221, 260–280. (IF = 4.6)
- 21 Peng Li, Tingting Li, Wai Sun Don, & **Bao-Shan Wang***. Scale-invariant multi-resolution alternative WENO scheme for the Euler equations. *Journal of Scientific Computing*, 2023, 94(1), 15. (IF = 2.5)
- 22 Yinghua Wang, Wai Sun Don, & **Bao-Shan Wang***. Fifth order AWENO finite difference scheme with adaptive numerical diffusion for Euler equations. *Computers & Fluids*, 2023, 251, 105743. (IF = 3.077)
- 23 Kang-Bo Tian, Wai Sun Don, & **Bao-Shan Wang***. High order WENO finite difference scheme with adaptive dual order ideal weights for hyperbolic conservation laws. *Applied Numerical Mathematics*, 2023, 187, 50–70. (IF = 2.994. 2023 年 10 月 25 日在过去 90 天最高下载文章中排第 16 名)
- 24 **Bao-Shan Wang**, Wai Sun Don, & Peng Li*. Fifth-order well-balanced positivity-preserving finite difference AWENO scheme with hydrostatic reconstruction for hyperbolic chemotaxis models. *Applied Numerical Mathematics*, 2023, 186, 41–56. (IF = 2.994)
- 25 **Bao-Shan Wang**, Wai Sun Don, Alexander Kurganov*, & Yongle Liu. Fifth-order A-WENO schemes based on the adaptive diffusion central-upwind Rankine-Hugoniot fluxes. *Communications on Applied Mathematics and Computation*, 2023, 5, 295–314. (IF = 1.6)

- 26 Yinghua Wang[#], **Bao-Shan Wang[#]**, Leevan Ling[#], & Wai Sun Don^{*:#}. A time-continuous embedding method for scalar hyperbolic conservation laws on manifolds. *Journal of Scientific Computing*, 2022, 93(3), 84. (IF = 2.5)
- 27 **Bao-Shan Wang**, Peng Li, & Zhen Gao^{*}. High order well-balanced and positivity-preserving finite difference AWENO scheme with hydrostatic reconstruction for shallow water equations. *Applied Numerical Mathematics*, 2022, 181, 483–502. (IF = 2.994)
- 28 **Bao-Shan Wang** & Wai Sun Don^{*}. Affine-invariant WENO weights and operator. *Applied Numerical Mathematics*, 2022, 181, 630–646. (IF = 2.994)
- 29 Wai Sun Don[#], Run Li[#], **Bao-Shan Wang[#]**, & Yinghua Wang^{*:#}. A novel and robust scale-invariant WENO scheme for hyperbolic conservation laws. *Journal of Computational Physics*, 2022, 448, 110724. (IF = 4.645)
- 30 Peng Li[#], **Bao-Shan Wang[#]**, & Wai Sun Don^{*:#}. Sensitivity parameter-independent characteristic-wise well-balanced finite volume WENO scheme for the Euler equations under gravitational fields. *Journal of Scientific Computing*, 2021, 88(2), 47. (IF = 2.843)
- 31 Zhen Gao^{*}, Qi Liu, Jan S. Hesthaven, **Bao-Shan Wang**, Wai Sun Don, & Xiao Wen. Non-intrusive reduced order modeling of convection dominated flows using artificial neural networks with application to Rayleigh-Taylor instability. *Communications in Computational Physics*, 2021, 30(1), 97–123. (IF = 3.246)
- 32 **Bao-Shan Wang**, Wai Sun Don, Naveen Kumar Garg^{*}, & Alexander Kurganov. Fifth-order A-WENO finite difference schemes based on a new adaptive diffusion central numerical flux. *SIAM Journal on Scientific Computing*, 2020, 42(6), A3932–A3956. (IF = 2.373)
- 33 Wai Sun Don, Dong-Mei Li, Zhen Gao, & **Bao-Shan Wang^{*}**. A characteristic-wise alternative WENO-Z finite difference scheme for solving the compressible multicomponent non-reactive flows in the over-estimated quasi-conservative form. *Journal of Scientific Computing*, 2020, 82(2), 27. (IF = 2.592)
- 34 Zhen Gao, Li-Li Fang, **Bao-Shan Wang**, Yinghua Wang, & Wai Sun Don^{*}. Seventh and ninth orders alternative WENO finite difference schemes for hyperbolic conservation laws. *Computers & Fluids*, 2020, 202, 104519. (IF = 3.013)
- 35 Peng Li, Xiqiang Zhao, Zhen Gao^{*}, & **Bao-Shan Wang**. High order hybrid weighted compact nonlinear schemes for hyperbolic conservation laws. *Advances in Applied Mathematics and Mechanics*, 2020, 12(4), 972–991. (IF = 1.727)
- 36 Yinghua Wang[#], **Bao-Shan Wang[#]**, & Wai Sun Don^{*:#}. Generalized sensitivity parameter free fifth order WENO finite difference scheme with Z-type weights. *Journal of Scientific Computing*, 2019, 81(3), 1329–1358. (IF = 2.228)
- 37 **Bao-Shan Wang**, Peng Li, Zhen Gao, & Wai Sun Don^{*}. An improved fifth order alternative WENO-Z finite difference scheme for hyperbolic conservation laws. *Journal of Computational Physics*, 2018, 374, 469–477. (IF = 2.845)
- 38 **Bao-Shan Wang**, Wai Sun Don^{*}, Zhen Gao, Yinghua Wang, & Xiao Wen. Hybrid Compact-WENO finite difference scheme with radial basis function based shock detection method for hyperbolic conservation laws. *SIAM Journal on Scientific Computing*, 2018, 40(6), A3699–A3714. (IF = 2.31)
- 39 Wai Sun Don[#], **Bao-Shan Wang[#]**, & Zhen Gao^{*:#}. Fast iterative adaptive multi-quadric radial basis function method for edges detection of piecewise functions—I: uniform mesh. *Journal of Scientific Computing*, 2018, 75(2), 1016–1039. (IF = 2.370)
- 40 王立锋^{*}, 叶文华, 陈竹, & 等. 激光聚变内爆流体不稳定性基础问题研究进展. *强激光与粒子束*, 2021, 33(1), 012001.

- 41 杨洋, 曾维新, 高振, & 王保山*. 求解双曲守恒律的紧-WENO 杂交格式及 RBF-FD 间断检测方法. 数值计算与计算机应用, 2020, 41(3), 232–245.
- 42 徐捷, 高振, 曾维新, & 王保山*. 高阶驻点上精度保持的 WENO 有限差分格式. 数值计算与计算机应用, 2020, 41(1), 68–82.
- 43 王保山, 曾维新, 高振*, & 陈玉虎. Compact-WENO 杂交格式中旋涡识别的多分辨率分析方法. 中国海洋大学学报 (自然科学版), 2018, 48(增 II), 198–202.

已接受论文

- 1 Yuanyang Qiao[#], Yun-Xia Liu[#], **Bao-Shan Wang**^{*,#}, & Wai Sun Don[#]. Affine-invariant trigonometric WENO finite difference schemes for hyperbolic conservation laws. *Communications in Computational Physics*, 2025. (IF = 3.1)

已投稿论文

- 1 Shaoshuai Chu, Alexander Kurganov*, Mingye Na, **Bao-Shan Wang**, & Ruixiao Xin. Local characteristic decomposition of equilibrium variables for hyperbolic systems of balance laws. submitted to *Applied Numerical Mathematics*. under revision.
- 2 Ya-Ru Zhao, Zhen Gao, & **Bao-Shan Wang***. Physical-constraints-preserving path-conservative Lax-Friedrichs schemes for non-conservative two-medium six-equation model. submitted to *Journal of Computational Physics*. under revision.
- 3 Qingcheng Fu, Alexander Kurganov*, & **Bao-Shan Wang**. Local characteristic decomposition-based central-upwind flux splitting. submitted to *Journal of Scientific Computing*. under revision.
- 4 Changming Guo[#], Yuanyang Qiao[#], **Bao-Shan Wang**[#], Wai Sun Don^{*,#}. Bound- and positivity-preserving first-order velocity-consistent HLL scheme for two-medium γ -based model of stiffened gas. in: D. E. Gildfind, R. Gollan and H. H. Kleine (eds), *Proceedings of the 35th International Symposium on Shock Waves: ISSW35*, under revision.
- 5 **Bao-Shan Wang***. On Wave Speeds in the HLLC Riemann Solver for the Compressible Two-medium Flows. submitted to *Journal of Computational Physics*. under revision.
- 6 Ya-Ru Zhao, Zhen Gao, **Bao-Shan Wang***, & Wai Sun Don. High-order physical-constraints-preserving common-weights A-WENO scheme for non-conservative five-equation model with stiffened gas EOS. submitted to *Journal of Computational Physics*. under revision.
- 7 Alex Shiu Lun Chu[#], Wai Sun Don[#], Leevan Ling[#], & **Bao-Shan Wang**^{*,#}. Foliation structures and global flow dynamics of scalar hyperbolic conservation laws on manifolds: II. An adaptively penalized closest point embedding with a WENO finite difference scheme (AP-cp-WENO). submitted to *SIAM Journal on Scientific Computing*.
- 8 Jia-Hao Liu, **Bao-Shan Wang**, & Ya-Ru Zhao*. Sixth-order AWENO schemes with adaptive linear weights for hyperbolic conservation laws. submitted to *Computers and Fluids*.
- 9 Changming Guo, Pengzhan Huang*, Yuanyang Qiao, **Bao-Shan Wang**. Accurate simulation of the granular avalanche flows by high-order positivity-preserving affine-invariant AWENO finite difference schemes. submitted to *Computers and Fluids*.
- 10 Yanping Qiu, Zhen Gao, Alexander Kurganov, **Bao-Shan Wang***, & Xiao Wen. Fifth-order well-balanced path-conservative A-WENO scheme for the Ripa model. submitted to *Journal of Computational Mathematics*.

- 11 Meifang Zhang[#], **Bao-Shan Wang**[#], Wai Sun Don[#], & Yuanyang Qiao^{*,#}. Spatially adaptive-order bound-preserving velocity-consistent AWENO schemes for compressible two-medium flows. submitted to *Journal of Scientific Computing*.
- 12 Hai-Liang Nie, Yuanyang Qiao, **Bao-Shan Wang**^{*}, & Wai Sun Don. RBF-AWENO scheme with affine-invariant weights: Achieving sixth-order accuracy and multiscale robustness. submitted to *Communications in Computational Physics*. under revision.
- 13 Naveen Kumar Garg^{*}, Jhantu Pal, & **Bao-Shan Wang**. A Low-Dissipation LLF-Type Adaptive Central Scheme for the Compressible Euler Equations. submitted to *Mathematics and Computers in Simulation*.
- 14 Jiayi Gu, **Bao-Shan Wang**^{*}, Wai Sun Don, & Jae-Hun Jung. A family of even-order central-upwind WENO schemes with averaged downwind and novel global smoothness indicators. submitted to *Numerical Methods for Partial Differential Equations*.
- 15 Jing Li, Peng Li^{*} & **Bao-Shan Wang**. Physical-constraints-preserving conservative-variable AWENO scheme for two-medium flows. submitted to *Advances in Applied Mathematics and Mechanics*.
- 16 邱燕萍, 高振, 王保山^{*}. 热旋转浅水波模型的五阶保平衡路径守恒特征 AWENO 格式. 已投稿至数值计算与计算机应用.

待整理论文

- 1 Shan Zhao, **Bao-Shan Wang**, & Peng Li^{*}. Bound-Preserving Central-Upwind Scheme for One- and Multi-Dimensional Detonation Simulations
- 2 Kang-Bo Tian, **Bao-Shan Wang**, & Zhen Gao^{*}. High-order well-balanced AWENO scheme for thermal rotating shallow water equations with Coriolis force.
- 3 Siyuan Lang, Zhiyue Zhang, **Bao-Shan Wang**, & Wai Sun Don. A physics-informed domain decomposition method with explicit shock tracking for 1D scalar conservation laws.
- 4 Xiao-Shuo Xiang, **Bao-Shan Wang**^{*}. Adaptive alternative weighted essentially non-oscillatory scheme for solving Degasperis-Procesi equation

待出版著作

- 1 Wai Sun Don & **Bao-Shan Wang**. Lecture on the High-Order WENO Scheme for Hyperbolic Conservation Laws (HKBU edition). DOI:10.13140/RG.2.2.24752.37123/2.
- 2 Wai Sun Don & **Bao-Shan Wang**. Comprehensive Lecture Notes on Hyperbolic Conservation Laws on Manifolds. DOI:
- 3 王保山. 我的 MATLAB 报告. 整理中.

研究项目

- | | |
|-------------------|--|
| 2024.01 – 2026.12 | ■ 国家自然科学基金青年项目/12301530/曲面双曲守恒律方程的时间连续最近点方法/30 万/主持 |
| 2025.01 – 2027.12 | ■ 中国海洋大学青年英才工程科研启动费/862501013106/极端条件下多介质流模型的高精度保物理约束数值方法研究/30 万/主持 |
| 2026.01 – 2028.12 | ■ 2025 年度山东省高等学校青年创新团队计划/多介质流模型数值模拟与应用创新团队/2025KJG007/多介质流模型保物理约束数值方法研究/主持 |

研究项目 (continued)

- 2023.01 – 2025.12 ■ 山东省自然科学基金面上项目/ZR2022MA012/可压缩多组分流模型基于物理结构的高精度自适应杂交求解器/10 万/负责/资助期满
- 2023.07 – 2024.07 ■ 中国博士后科学基金面上项目/2023M733348/Mie-Grüneisen 状态方程下两介质六方程模型的保界限制器研究/8 万/主持/结题
- 2024.01 – 2027.12 ■ 国家自然科学基金面上项目/12371435/基于深度神经网络的新型变分同化方法研究/43.5 万/参与/第五位
- 2024.01 – 2026.12 ■ 山东省自然科学基金面上项目/ZR2023MA077/反常扩散方程的局部误差估计与误差界爆破现象研究/10 万/参与/第二位
- 2024.01 – 2026.12 ■ 山东省自然科学基金面上项目/ZR2023MA043/数据驱动的南海内波高效数值模拟方法研究/10 万/参与/第五位
- 2024.01 – 2026.12 ■ 山东省高等学校青年创新团队发展计划/2023KJ1020/海洋内波的数学分析与求解理论研究/0 万/参与/第六位
- 2022.10 – 2025.09 ■ 崂山实验室创新项目/LSKJ202202304/北极海冰预测系统的国产计算平台移植与优化/300 万/参与/第十位/结题
- 2019.01 – 2022.12 ■ 国家自然科学基金面上项目/11871443/不确定性下的三维气体爆炸波的数值方法研究/50 万/参与/第八位/结题
- 2017.01 – 2019.12 ■ 国家重大科技专项项目/****/基础问题研究/90 万/参与/第八位/结题

教学经历

- 2026.03 – 2025.07 ■ 本科生课程《复变函数》(代码 008401101031, 100 + 100 人)
- 2026.03 – 2025.07 ■ 研究生课程《谱方法》(代码 100K0171001, 3 人 + 2 人旁听)
- 2025.02 – 2025.07 ■ 本科生课程《复变函数》(代码 008401101031, 149 + 76 人)
- 2025.02 – 2025.07 ■ 研究生课程《谱方法》(代码 100K0171001, 6 人 + 2 人旁听)
- 2021.09 – 2022.01 ■ 迷你课程:《科学计算与编程》、《非线性波动方程》
- 2020.03 – 2020.07 ■ 研究生课程《谱方法》

学术活动

- 2026.08.27 ■ Foliation Structures and Global Flow Dynamics of Scalar Hyperbolic Conservation Laws on Manifolds, EASIAM 2026, Jeju, South Korea
- 2026.08.18 ■ High-order physical-constraints-preserving velocity-consistent schemes for compressible multicomponent five-equation model with the Mie-Grüneisen equation of state, 昆明理工大学, 昆明, 云南.
- 2026.01.30 ■ 浅水波类模型的高精度保平衡 AWENO 格式, 崂山国家实验室, 青岛
- 2026.01.17 ■ Foliation Structures and Global Flow Dynamics of Scalar Hyperbolic Conservation Laws on Manifolds, 偏微分方程高精度数值算法及应用研讨会, 广西大学, 南宁
- 2026.01.07 ■ —, Conference on Modern Kernel Methods and Applications, On the Occasion of Robert Schaback's 80th Birthday, HKBU
- 2025.10.26 ■ A family of bound-preserving velocity-consistent schemes for two-medium γ -based model with stiffened gas, CSIAM 2025 年会, 长沙, 湖南
- 2025.08.09 ■ —, 第二十一届全国流体力学数值方法研讨会, 伊宁, 伊犁
- 2025.07.27 ■ Bound- and positivity-preserving path-conservative central-ipwind AWENO scheme for the five-equation model of compressible two-component flows, 新疆大学, 乌鲁木齐

学术活动 (continued)

- 2024.12.31 ■ 一，河南工业大学，郑州
- 2024.11.11 ■ A Time-Continuous Embedding Method for Hyperbolic Conservation Laws on Manifolds, 香港浸会大学数学系，香港
- 2024.06.24 ■ Fifth-order bound-, positivity-, and equilibrium-preserving affine-invariant AWENO scheme for two-medium γ -based model of stiffened gas, 新疆大学，乌鲁木齐
- 2024.04.07 ■ Time-Continuous Embedding Method for Scalar Hyperbolic Conservation Laws on Manifolds, 石家庄铁道大学，石家庄
- 2024.03.19 ■ Affine-invariant WENO operator and applications in solving hyperbolic conservation laws, Workshop on Development of High-Order Methods for Hyperbolic PDEs, 南方科技大学杰曼诺夫数学中心，深圳
- 2023.11.11 ■ Time-Continuous Embedding Method for Scalar Hyperbolic Conservation Laws on Manifolds, 哈尔滨工业大学(深圳)，深圳
- 2023.10.24 ■ Affine-invariant WENO weights and applications in solving hyperbolic conservation laws, 香港浸会大学，香港
- 2023.08.16 ■ 一，ICOSAHOM 2023，首尔
- 2023.04.04 ■ 一，杰曼诺夫数学中心，深圳
- 2023.04.02 ■ 一，第二十届全国流体力学数值方法研讨会，江苏南京
- 2022.07.12 ■ 仿射不变性的高精度低耗散 WENO 格式，新疆大学（线上）
- 2021.07.03 ■ Alternative WENO Scheme with Adaptive Diffusion Numerical Flux for Hyperbolic Conservation Laws, 山东省计算数学学术研讨会，山东烟台
- 2020.11.15 ■ 一，中国工业与应用数学学会学生论坛（线上）
- 2019.11.05 ■ Non-intrusive reduced order modeling of convection dominated flows using artificial neural networks with application to Rayleigh-Taylor instability. International Joint Conference on AI & Data Science: Mathematics & Applications, 韩国亚洲大学
- 2019.06.24 ■ Hybrid compact-WENO finite difference scheme with radial basis function based shock detection method for hyperbolic conservation laws. The 11th International Conference on Computational Physics, 浙江大学
- 2019.04.20 ■ 一，石家庄铁道大学
- 2018.12.07 ■ 烧蚀瑞利泰勒不稳定性数值模拟，北京应用物理与计算数学研究所
- 2017.08.09 ■ Hybrid compact-WENO finite difference scheme with radial basis function based shock detection method for hyperbolic conservation laws, 第十八届全国流体力学数值方法研讨会，怀化学院

专业服务

期刊审稿 (部分列表)

- 29 份 ■ Journal of Computational Physics
- 26 份 ■ Applied Numerical Mathematics
- 12 份 ■ Advances in Applied Mathematics and Mechanics
- 6 份 ■ Communications in Nonlinear Science and Numerical Simulation
- 6 份 ■ Computers & Fluids
- 5 份 ■ Journal of Computational and Applied Mathematics
- 4 份 ■ Computers and Mathematics with Applications

专业服务 (continued)

- 2 份 ■ Advances in Water Resources
- 1 份 ■ Communications in Computational Physics
- 19 份 ■ 其他期刊

专题研讨会组织者

- 2026 ■ EASIAM 2026, High-Order Numerical Methods for Hyperbolic Conservation Laws, 8 月 23–27 日, 韩国济州岛
- 2023 ■ ICOSAHOM 2023, Recent developments in physical-property-preserving WENO schemes and DG methods for hyperbolic single- and multi-medium models, 8 月 14–18 日, 延世大学, 韩国首尔

学术兼职

- 2025– ■ 骨干成员, 山东省随机系统控制理论与科学计算重点实验室
- 2024– ■ 骨干成员, 海洋数学技术联合实验室
- 2022– ■ 专业会员, 中国工业与应用数学学会

获奖情况

- 2023 ■ 山东省优秀博士学位论文
- 2021 ■ 博士研究生国家奖学金、中国海洋大学第八届研究生年度人物称号 (共 10 项)
- 2019 ■ 山东省优秀硕士学位论文
- 2018 ■ 山东省优秀硕士毕业研究生
- 2017 ■ 硕士研究生国家奖学金
- 2016 ■ 研究生数学建模竞赛国家一等奖和赛事优秀论文
- 2015 ■ 研究生数学建模竞赛国家二等奖
- 2014 ■ 大学生数学建模竞赛国家二等奖

其他技能

- 创建微信公众号“台劳公式”: 分享科研经验、编程技术和快乐数学 (3200+ 关注)
- 熟练掌握软件: MATLAB 和 Fortran 编程, LaTeX 文档排版
- 熟练使用 Linux 操作系统, 维护 Linux 服务器

指导学生情况

博士研究生 (协助指导)

- 2025 – 2029 ■ 赵亚茹 [在读]; 郭长明 (新疆大学) [在读]
- 2022 – 2027 ■ 项晓硕 [在读]
- 2021 – 2023 ■ 王海云, 可压缩多介质流数值模拟的保界保正间断 Galerkin 方法研究, 湘潭大学
- 2020 – 2026 ■ 田康博, 三类浅水波方程五阶保平衡保正 WENO 有限差分格式的研究, 青岛黄海学院
- 2020 – 2025 ■ 任佰芬, 浅水波方程组的保界保平衡高精度有限差分格式研究, 山东工商学院

指导学生情况 (continued)

硕士研究生

- 2026 – 2029 ■ 丁佳旭, 赵杉 [在读]
2025 – 2028 ■ 刘家豪 [在读]

硕士研究生 (协助指导)

- 2024 – 2027 ■ 聂海亮 (新疆大学) [在读]; 张美芳 (新疆大学) [在读]; 杨鑫云 (新疆大学) [在读];
 ■ 李晶 (石家庄铁道大学) [在读]
2023 – 2026 ■ 邱燕萍 [毕业]; 赵亚茹 [2025 年硕博连续] 郭长明 (新疆大学) [2026 年硕博连续]
2022 – 2025 ■ 宋宜璇 (新疆大学) [毕业]; 汤子钰 [毕业]
2021 – 2024 ■ 王彩凤 [德州审计局]; 李嘉乐 [德州统计局]; 郭爽 [泰安市中行];
 ■ 刘馨茹 [连云港税务局]; 刘云霞 (新疆大学) [北京某科研机构]
2020 – 2023 ■ 付青城, 获硕士国家奖学金、校优秀硕士学位论文, 去南方科技大学数学系读博,
 师从 Alexander Kurganov 教授
2019 – 2022 ■ 李闰, 获硕士国家奖学金, [重庆行知学校];
 ■ 李婷婷 (石家庄铁道大学) [北京通州运河中学]
2017 – 2020 ■ 李东梅 [青岛体育运动学校]; 方丽丽 [滨州财政局]; 徐捷 [舟山中学]; 杨洋 [毕业]

本科生

- 2021 – 2025 ■ 俞磊, 留在学院读研, 师从付凯副教授; 于子峻

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